
GHZ States on Real QPUs: an Ion / Superconducting Trilogy, and a 97.27% Four-Ion State

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Hardware: Rigetti Cepheus-1-108Q, IQM Garnet, AQT IBEX Q1 via Open Quantum (attribution: www.openquantum.com/citation)

Abstract

Fourteen jobs were executed on 29/09/2026 on three real quantum processors. The centerpiece is a four-qubit GHZ state on trapped ions (job `df23deac`) measured at 97.27%, i.e. $(497 + 499)/1024$ shots. The GHZ-4 trilogy ranks ions $97.3\% > \text{Garnet } 94.6\% > \text{Cepheus } 79.3\%$. Bell states reach 99.61% and GHZ-7 90.04% on IBEX; a twelve-qubit GHZ ("chat-12") reaches 66.0% on Garnet. Measured costs are 1 / 2 / 15 Spark credits per 1024 shots for Cepheus / Garnet / IBEX. The ion slope is compatible with -1.9 fidelity points per qubit, with local slopes -1.2 then -2.4 : a decay that accelerates.

Scientific question

How does GHZ fidelity decay with qubit count, and how does that decay differ between trapped-ion and superconducting modalities?

Method

Multi-line QASM (`include "qelib1.inc";`), job subcategory `phys:oth`, quote approved only after reading the device quote line ("Auto-selected plan: N credits"). Laboratory doctrine: one shot batch at a time, order Garnet → Cepheus → IBEX; never cancel a job without an explicit order; access tokens have a ≈ 5 min TTL and are refreshed per call.

Results

- GHZ-4 trapped ions, `df23deac`: 97.27% — counts $0000 = 497$, $1111 = 499$ over 1024 shots.
- GHZ-4 trilogy: ions $97.3\% > \text{Garnet } 94.6\% > \text{Cepheus } 79.3\%$.
- Bell 99.61% and GHZ-7 90.04% on IBEX Q1 (15 Spark each).
- Separated GHZ-3/4/5 on Garnet: 94.5 / 94.6 / 88.5% (2 Spark); Cepheus trilogy 89.7 / 79.3 / 68.7% (1 Spark).
- Chat-12 (GHZ-12, `b373d22f`): 66.0% (2 Spark).
- Ion slope ≈ -1.9 pt/qubit over three points (local slopes -1.2 then -2.4).

The raw counts of `df23deac` were re-downloaded and recomputed on 03/10/2026 from the signed output URL: 97.27% confirmed identically.

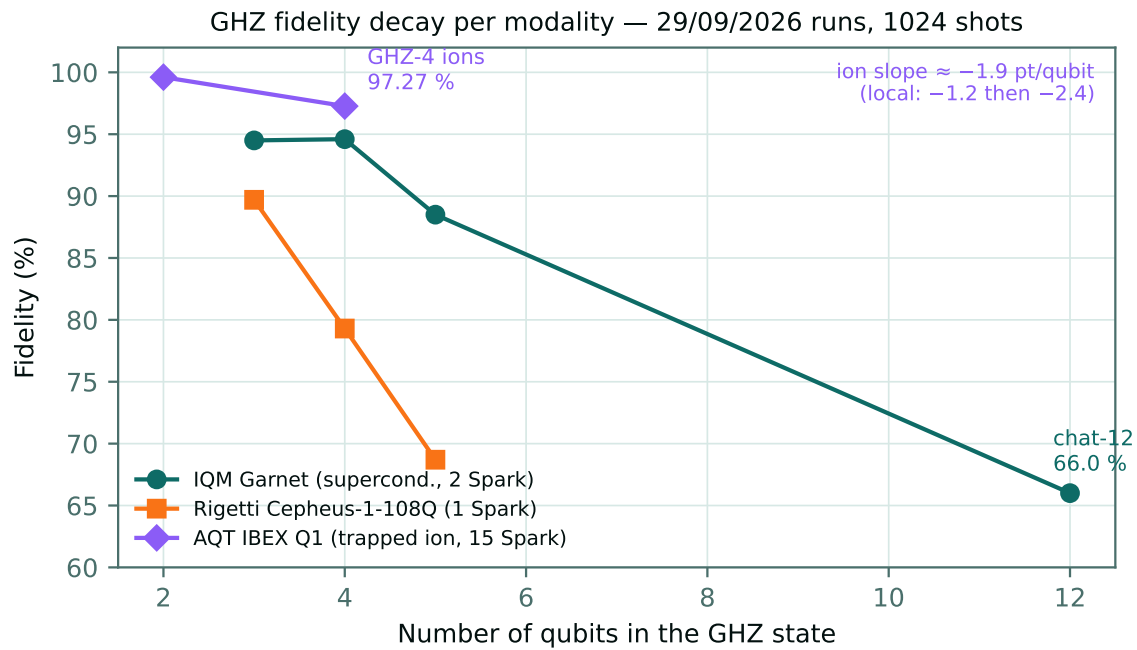


Figure 1: GHZ fidelity versus qubit count per modality (29/09/2026, 1024 shots). Ion points: Bell 99.61% (2q), GHZ-4 97.27%. Garnet: 94.5 / 94.6 / 88.5 / 66.0%. Cepheus: 89.7 / 79.3 / 68.7%.

Negative results

- Episodic-shot runs on co-hosted Garnet collapsed to 45% vs 94% → RATISS law of the episodic shot: all-to-all ONLY on co-hosted devices.
- Job 407cd969 cancelled → 15 Spark refunded (first refund of the lab).
- Chat-12 at 66.0%: below target, published as is.

Controls

Quote read before approval; raw counts archived; server timestamps replayable. Known SDK defect: `download_job_output` raises 'str' object has no attribute 'output_data_url' (still present in SDK 0.4.3); harvesting uses `get_job(...).output_data_url` instead.

Reproducibility

Raw counts: `RATISS-PLANCK/resultats/qpu_ghz4_ibex_counts.json`; register of the 14 jobs: `RATISS-ARCHIVES/preuves`

Limitations

1024 shots: statistical uncertainty not reduced further (credit cost). Cross-modality comparison is not at constant qubit count beyond the trilogy.

Data and code availability

<https://github.com/jonathanssearch/RATISS-PLANCK> (seal 52/52, MIT). Platform attribution mandatory: www.openquantum.com/citation. Zenodo DOI: to be assigned.

RATISS Labs — independent, single-author laboratory, Yaoundé (Cameroon). No number in this document is invented: every value carries a source in Section “Data and code availability” or in the references. Negative results are published with the same prominence as positive ones.